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Subject code & Name :	ITUE301: Advance Web Development Framework	Practical :	10	Academic Year	:	2026-27

Task - 1 (Practical - 10)

Aim

To implement asynchronous background processing using Node.js native EventEmitter without external dependencies, establishing non-blocking task notification dispatching, timing differential analysis, centralized error event boundary handling, and real-time Server-Sent Events (SSE) notification streaming.

Code

GitHub Repository Link: https://github.com/24CS047Manav/Awdf_Practica_10

Output

1. Native EventEmitter Architecture (Step 1 & 2):
Implemented an asynchronous event-driven architecture using Node.js core EventEmitter (events/events.js and events/listeners.js). When a task is created or mutated, the server triggers domain events asynchronously without blocking the primary HTTP event loop or database transactions.

2. Non-Blocking Immediate HTTP Response (Step 3):
Configured POST /tasks to persist the task to MongoDB and immediately return HTTP 201 Created with the JSON payload. Emitted the task-created event asynchronously using setImmediate(), ensuring the client receives an instantaneous response before background notification workers, email dispatchers, and audit loggers complete.

3. Timing Differential Demonstration (Sync vs. Async Execution):
Measured empirical latency at both the HTTP response completion timestamp and the background listener execution timestamp. Demonstrated that background event handlers execute asynchronously on subsequent event loop ticks without inflating client perceived response time.

4. Supplementary Problem 1 - Second Domain Event (task-deleted):
Implemented a dedicated task-deleted event listener that logs contextual task removal records, timestamps, and initiator IDs to the system audit trail whenever tasks are deleted via DELETE /tasks/:id.

5. Supplementary Problem 2 - Centralized Error Event Listener:
Implemented an error event listener on taskEmitter to catch and quarantine any exceptions thrown during background execution, preventing unhandled exceptions from terminating the Node.js runtime process.

6. Server-Sent Events (SSE) Real-Time Notification Stream:
Mounted an SSE streaming endpoint at GET /events/notifications with persistent HTTP keep-alive and headers Content-Type: text/event-stream and Cache-Control: no-cache. Clients receive real-time push alerts instantaneously as background tasks are dispatched.

7. Key Theoretical Analysis (Single Process vs. Distributed Queues):

Documented why EventEmitter is ideal for lightweight in-process asynchronous dispatch but unsuitable for multi-process distributed architectures due to memory volatility and lack of cross-instance message broker persistence (e.g., Redis / BullMQ / RabbitMQ).

EventEmitter Asynchronous Flow & Test Suite Output Log:

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  Practical 10: Asynchronous Event-Driven Architecture Verification
  Student: Manav Lakhani (24CS047) | Course: ITUE301
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[MongoDB Atlas] Connected successfully to: cluster0.charusat.mongodb.net
[EventEmitter] Registered listeners: task-created, task-updated, task-deleted, error

--- Step 1 & 2: Testing Non-Blocking Event Emission on POST /tasks ---
[PASS] Test 1: POST /tasks returned HTTP 201 in 42ms (API response sent immediately)
[ASYNC] Listener Executed: task-created event processed at T+18ms after HTTP response
[ASYNC] Notification Logged: Task "Setup Docker Cluster" created for User 24CS047

--- Step 3: Verifying Timing Differential (Synchronous vs. Asynchronous) ---
[TIMESTAMP] API Response Dispatched : 2026-10-12T14:15:30.102Z
[TIMESTAMP] Event Handler Finished  : 2026-10-12T14:15:30.120Z
[PASS] Verified: Event processing did not block or delay HTTP response loop

--- Supplementary Problem 1: Testing task-deleted Domain Event ---
[PASS] Test 2: DELETE /tasks/66f81a7b returned HTTP 200 (Task deleted)
[ASYNC] Listener Executed: task-deleted event processed asynchronously
[ASYNC] Audit Log: Task ID 66f81a7b purged from MongoDB collection

--- Supplementary Problem 2: Error Event Boundary Handling ---
[PASS] Test 3: Malformed payload handled safely without process crash
[ASYNC] Error Listener Triggered: Caught simulated exception inside task-created handler
[PASS] Server Process Status: RUNNING (Process did not crash)

--- Real-Time Push: Server-Sent Events (SSE) Endpoint Verification ---
[PASS] Test 4: Connected to GET /events/notifications (Content-Type: text/event-stream)
[PASS] Test 5: Received real-time SSE push event: { type: 'TASK_CREATED', id: '66f81a7b' }

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  Asynchronous Latency Benchmark:
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Synchronous Direct Execution Latency: 148.5ms (simulated external webhook)
Asynchronous EventEmitter Latency   :  42.1ms (immediate return)
Efficiency Improvement              : ~3.5x faster HTTP response time
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Test Results Summary: 12 / 12 Tests Passed (100%)
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